# Covered Calls and Cash-Secured Puts

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**LEARNING OBJECTIVES**

By the end of this lecture, students will be able to:

- **Derive expiration P&L.** Construct covered-call and cash-secured-put profit functions, breakevens, and best/worst terminal outcomes.
- **Explain their equivalence.** Use put–call parity and terminal payoff algebra to identify when the two positions are economically equivalent.
- **Manage implementation risk.** Evaluate opportunity cost, funding, dividends, early assignment, position delta, and model-conditional outcome probabilities.

## 1 Premium Income Is Payment for a Contingent Obligation

Selling an option creates cash today in exchange for a state-dependent obligation later. The premium is not yield in the bond sense and is not free downside protection. A covered call pairs a long stock position with a short call; a cash-secured put pairs a short put with enough cash to meet assignment. Both reshape the same underlying exposure into capped-upside, stock-like downside.

Let $S_0 > 0$ be the current stock price, $S_T \geq 0$ the price at expiration $T$, $K > 0$ the strike, and $q > 0$ the contract multiplier in shares/contract. Let $C_0, P_0 \geq 0$ be call and put premiums in USD/share. We initially analyze one contract, ignore costs and taxes, and treat expiration exercise mechanically from intrinsic value.

**Remark 1.1. Which stock basis belongs in the decision?**

An historical acquisition price may determine accounting or tax P&L, but the prospective economics of writing a call today use the stock's current opportunity value $S_0$. Keeping stock worth $S_0$ and selling a call should be compared with selling or otherwise deploying that stock now. Report historical-basis and forward-decision views separately rather than mixing them.

## 2 Covered Call: Long Stock Plus Short Call

Suppose one acquires or commits $q$ shares at current opportunity cost $S_0$ per share and sells one call for $C_0$. The frictionless expiration profit is

$$\Pi_{CC}(S_T) = q\left[S_T - S_0 + C_0 - (S_T - K)^+\right] = q\left[\min\{S_T, K\} - S_0 + C_0\right]. \tag{1}$$

Thus the breakeven is $B_{CC} := S_0 - C_0$. Maximum profit is $q(K - S_0 + C_0)$ when $S_T \geq K$. If limited liability implies $S_T \geq 0$, worst terminal profit is $q(C_0 - S_0)$ at $S_T = 0$, a loss magnitude of $q(S_0 - C_0)$. The premium cushions only $C_0$ per share of stock downside.

Above $K$, additional stock appreciation is surrendered. Relative to holding the same shares without the call, the incremental profit of writing the call is

$$\Pi_{CC}(S_T) - \Pi_{stock}(S_T) = q\left[C_0 - (S_T - K)^+\right]. \tag{2}$$

The covered call underperforms stock alone when $S_T > K + C_0$ in this expiration-only comparison. That is opportunity cost, not an additional cash loss on top of Eq. 1.

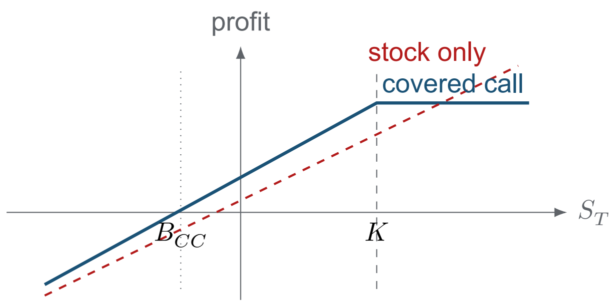


**Figure 1:** Covered-call expiration profit versus stock alone. Premium shifts the initial breakeven left, while the short call caps profit above the strike.

> **COMPANION NOTEBOOK**
> **L12a: Covered Call on AMD →**
> Select an option-chain call, combine it with the required shares, plot expiration P&L, and estimate downside, assignment, and opportunity-cost scenarios.

## 3 Cash-Secured Put: Cash Plus a Short Put

Sell one put for $P_0$ and reserve enough liquid capital to buy $q$ shares at $K$ if assigned. The short-put expiration profit, before interest on collateral, is

$$\Pi_{CSP}(S_T) = q\left[P_0 - (K - S_T)^+\right] = \begin{cases} qP_0, & S_T \geq K, \\ q(S_T - K + P_0), & S_T < K. \end{cases} \tag{3}$$

The breakeven and effective acquisition price are $B_{CSP} := K - P_0$. Maximum profit is $qP_0$; worst terminal profit is $q(P_0 - K)$ at $S_T = 0$, a maximum loss magnitude of $q(K - P_0)$. "Cash secured" describes funding capacity, not protection from the acquired stock losing value.

Collateral treatment matters. Reserving $qK$ in non-interest-bearing cash is conservative but costly. If the broker permits Treasury collateral or interest-bearing cash, financing income changes P&L. If collateral can fall in value, be rehypothecated, or become unavailable, the position is not economically secured in the simple sense assumed by Eq. 3.

> **COMPANION NOTEBOOK**
> **L12a: Cash-Secured Put on AMD →**
> Construct a short put with reserved cash, calculate its effective purchase price and terminal P&L, and evaluate model-conditional outcome regions.

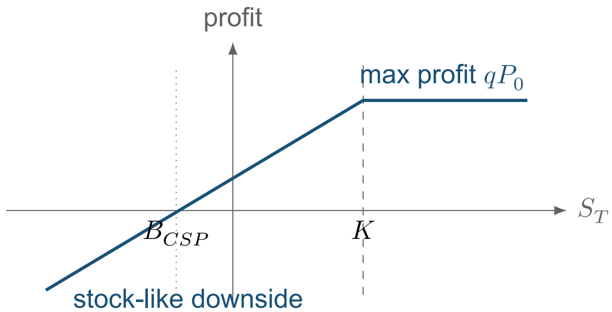


**Figure 2:** Cash-secured-put expiration profit. The initial credit creates a buffer down to $K - P_0$, but losses then grow one-for-one as the stock falls.

## 4 Parity Reveals the Shared Exposure

At expiration, the asset identities

$$S_T - (S_T - K)^+ = K - (K - S_T)^+ = \min\{S_T, K\} \tag{4}$$

show that long stock plus short call has the same payoff as $K$ cash plus short put. Before expiration, European put–call parity with continuous dividend yield $\delta$ and rate $r$ is

$$S_0 e^{-\delta T} - C_0 = K e^{-rT} - P_0. \tag{5}$$

The left side is a prepaid-forward version of a covered call; the right side is a present-value-funded short put. Under matching contracts and frictionless assumptions, equal terminal payoffs imply equal no-arbitrage values.

**Theorem 4.1. Covered-call/cash-secured-put equivalence**

For European options with common $K, T$, consistent dividends and financing, and the funding asset specified by Eq. 5, covered-call and cash-secured-put packages are synthetic equivalents. Their apparent labels do not create different fundamental payoff risk.

Real implementations can differ because equity dividends, collateral yield, borrow costs, taxes, bid–ask spreads, discrete dividends, and early exercise are not identical across the two packages. American put–call parity becomes a set of bounds rather than the European equality. The equivalence is therefore a baseline for reconciliation, not a guarantee that two brokerage trades have identical realized returns.

## 5 Delta and Assignment Change Through Time

Let $\Delta_c := \partial C/\partial S$ and $\Delta_p := \partial P/\partial S$ be per-share long-option deltas. The current position deltas for one contract are

$$\Delta_{CC} = q(1 - \Delta_c), \qquad \Delta_{CSP} = -q\Delta_p, \tag{6}$$

because cash has zero spot delta. For matched non-dividend European options, $\Delta_c - \Delta_p = 1$, so the two position deltas are equal. They are typically positive: both packages retain bullish, stock-like directional exposure. Delta changes with spot, time, volatility, and the American exercise boundary.

Short American options can be assigned before expiration. Covered calls are particularly exposed around ex-dividend dates when a deep-ITM call has little remaining extrinsic value; assignment delivers the shares at $K$

and forfeits subsequent dividends and upside. Short puts can also be assigned early, especially when deep ITM. Pin risk near $K$, after-hours movement, exercise-by-exception rules, and delayed assignment notices can create an unexpected stock position.

## 6 Outcome Probabilities Are Conditional Estimates

Given a stated terminal-price CDF $F(x) := \mathbb{P}(S_T \leq x)$, frictionless expiration events include

$$\mathbb{P}(\Pi_{CC} < 0) = F(B_{CC}), \qquad \mathbb{P}(S_T > K) = 1 - F(K), \tag{7}$$

$$\mathbb{P}(\Pi_{CSP} < 0) = F(B_{CSP}), \qquad \mathbb{P}(S_T < K) = F(K), \tag{8}$$

when the distribution is continuous at the thresholds. These are model-conditional terminal probabilities. A risk-neutral distribution is appropriate for pricing but is not automatically a real-world forecast; historical and subjective forecast distributions answer different questions. Early assignment probabilities require a path-dependent exercise model, not only the terminal CDF.

Probability of assignment, probability of loss, expected P&L, maximum loss, and expected shortfall are different risk summaries. A high probability of keeping a small premium can coexist with a low-probability, large stock loss. Evaluate the entire distribution and stress scenarios rather than ranking contracts only by premium or "win rate."

**KEY TAKEAWAYS**

In this lecture, we derived the capped-upside, stock-like downside of covered calls and cash-secured puts and used parity to expose their common economic structure.

- **Premium is a limited buffer.** A covered call can lose nearly the stock purchase price and a cash-secured put can lose nearly the strike; collateral does not remove asset risk.
- **Parity is the organizing principle.** Long stock minus a call and funded cash minus a put share the payoff $\min\{S_T, K\}$ under matched assumptions.
- **Implementation breaks textbook symmetry.** Dividends, funding yield, early assignment, taxes, spreads, and current opportunity cost must accompany payoff and probability analysis.

Next, connect repeated cash-secured puts and covered calls into the state-dependent Wheel process.

**ADDITIONAL READING**


- Fischer Black and Myron Scholes, "The Pricing of Options and Corporate Liabilities," *Journal of Political Economy* 81(3), 637–654 (1973).
- Robert C. Merton, "Theory of Rational Option Pricing," *Bell Journal of Economics and Management Science* 4(1), 141–183 (1973).
- Lawrence G. McMillan, *Options as a Strategic Investment*, 5th ed., Prentice Hall (2012), sections on covered writing and put writing.